NEXUS-1 — Project Brief

Nuclear Plant Operator Console & Digital-Twin Platform · one-look overview · Phase 0

An interactive operator console and digital-twin demonstration for a nuclear power facility, delivered as a single self-contained web application driven by a live in-browser simulation. It presents an entire plant - physics, generation, safety, fleet, lifecycle, people and analytics - through one navigable interface, designed as a connected system rather than a set of separate dashboards.

At a glance

Reactor core map, control rods + SCRAM, neutronics, real point-kinetics, 3D twin	**Fleet** multi-unit PWR + SMR modules, grid aggregation, unit selector
Secondary & electrical steam generators, coolant loops, power & grid sync	**Safety & monitoring** radiation, alarms, incidents, hash-sealed audit, OT security
Intelligence AI diagnostics, RL optimiser (advisory), dependency graph/matrix/causal chain, trends	**Integrity & lifecycle** NDT rod inspection, component ageing, decommissioning
People, robots & access staffing + stress test, robotic fleet + readiness, zone access	**Connected behaviour** a SCRAM ripples into incidents, compliance, inspection & ageing

What is real vs. simulated

Real / genuine	Simulated / illustrative
Interface, navigation, visualisations and interactions are working code.	All data is generated in the browser; nothing connects to a real plant.
The point-kinetics physics and the RL agent are real in structure and genuinely compute/learn.	Their constants are representative, not tuned to a specific reactor.
Workflows reflect real industry concepts (NDT, dose limits, decommissioning stages).	Ageing figures, weights, names, ratings and dates are plausible placeholders.

Destination & technology

Goal: a platform active in all modes — Training, Shadow, Advisory, Supervisory and Hybrid Digital Twin — built out from this Phase 0 Simulation baseline. It is a monitoring/analytics/advisory layer with **no control authority**; any real-plant link is read-only until a formal assurance and certification path is completed.

Stack (production target): .NET C# microservices (no Orleans) · Angular · SignalR · SQL Server · Azure Service Bus + Event Hubs · Docker/Kubernetes · OpenTelemetry.

Code: github.com/gregory82gr/Nexus-1-phase-0 | **Live demo:** gregory82gr.github.io/Nexus-1-phase-0

By **Grigorios Agathangelidis** (Electrical & Software Engineering) - a personal project applying software-architecture, simulation, visualisation and systems-design skills to a studied domain; not a claim of nuclear-engineering expertise. Demonstration and educational tool; not a licensed analysis tool and not connected to any real reactor. Build NX1-094C79E0A366D34F.